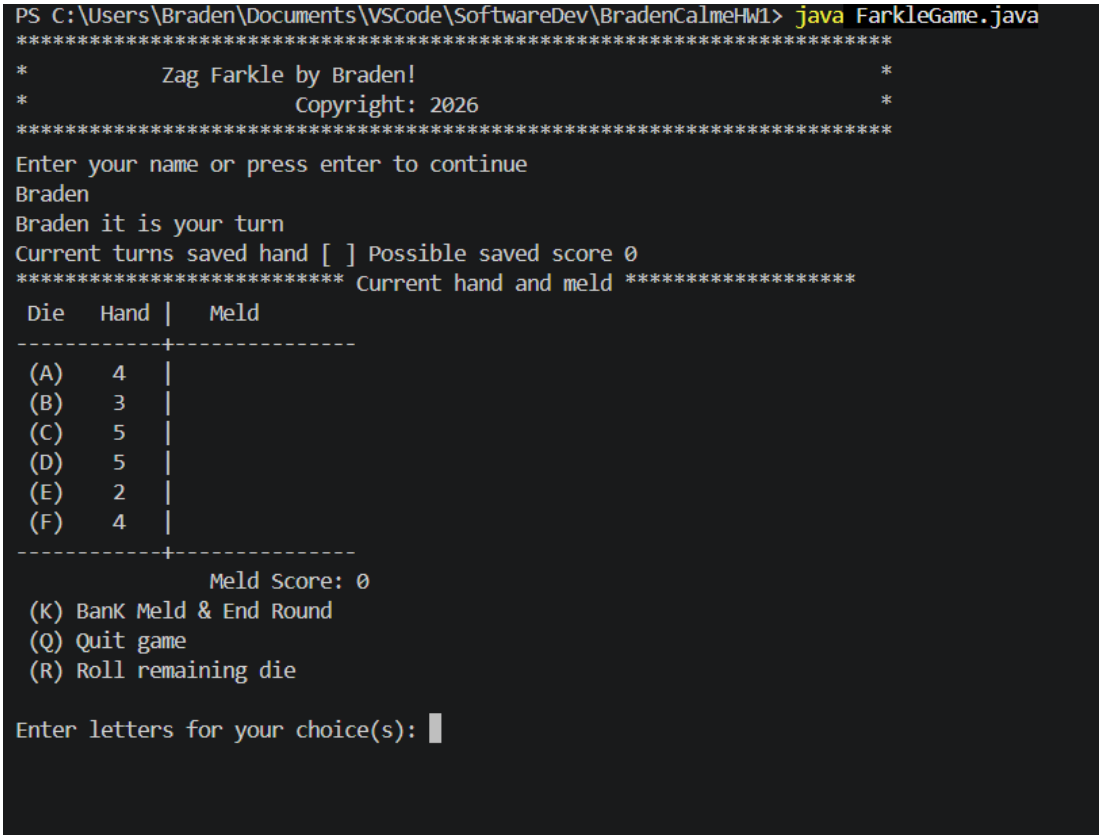
A summary of the goal or purpose of the program in your own words.

The program is designed to fully play a single turn in Farkle. A player is able to create a meld and then bank that melds score. They are able to reroll dice as long as a valid meld is present. Game will automatically display a farkle if present. User is now able to enter there name and will be addressed when it is their turn. Added a hot hands feature that now sees if a player has rolled all six dice as valid numbers and wants to continue their turn.

An overview of the general design you chose for your program.

I implemented a valid meld counter to see if they had rolled six valid dice. I split the program into a player, score calculator, dice, farkle game, and saved meld class. The farkle game is what runs the actual turn and uses the other classes for their specific purposes. Score calculator keeps the meld score up to date. Player stores the name and total score a player has accumulated. Dice stores the unmelded die values and creates new rolls. Saved meld stores the saved die that are not being rerolled in the turn.

A screenshot of your program's starting screen with your intro banner and first meld menu.



A short description of one of your unit tests and how that test demonstrates a correct behavior for your project.

The test straight test creates a meld with a valid straight among the die values. It then calculates the points for that meld and runs an if statement to see if the correct point value is given. If it is correct, it will return true.

A description of any major design and/or programming issues, why these were the major issues, how you addressed them, and why you addressed them the way you did.

I was struggling with how I was supposed to create the test melds. I than learned about the varargs parameter which basically takes a series of values and puts them in an array. This was efficient and made it easy to create new melds. I also struggled with calculating a hot hand. I decided to go with a valid die count because it seemed easiest and did not require a complete rework of my code.

A retrospective of what you would have done differently if you had more time

I might have made another class for the take turn function as it has become quite complicated. I feel like I might also have a couple repeating variables basically storing the same values so I would clean that up.